

		$v_{\text{esc}} = \sqrt{R}$
13	C	$\sqrt{\langle c^2 \rangle} = \sqrt{\frac{3RT}{M_m}}$ $= \sqrt{\frac{3(8.31)(273.15 + 30)}{4.0 \times 10^{-3}}}$ $\approx 1400 \text{ m s}^{-1}$
14	C	The internal energy U , of a system, is the sum of a random distribution of kinetic and